Review of "Z-residuals for Diagnosing Bayesian Models"

1 Summary

The manuscript deals with the problem of diagnosing Bayesian models using randomised survival probabilities (RSPs) based on full posterior draws, as well as leave-one-out (LOO) posterior draws. The authors develop theoretical guarantees relating to the asymptotic uniformity of these RSPs under both full posterior and LOO posterior draws, and present some numerical demonstrations on simulated and real data demonstrating their promise.

I will quickly give some context on my own expertise, and thus my interests and limitations as a reviewer. My work has primarily focussed on understanding the theoretical characteristics of Bayesian prediction, prior elicitation, and model selection. While I have come across some of the methods discussed in this manuscript, and some other literature which I believe to be related but unreferenced by the authors, my own related expertise lies in using cross-validation for the same ambitions. As such, I apologise in advance should some of my comments be misguided.

1.1 Overall evaluation

Bayesian model diagnostics, and in particular predictive diagnostics, are becoming increasingly important in our field. And many of the approaches I have seen used in practice exist without the theoretical guarantees the authors have developed here. As such, I believe the ambition of this manuscript to be very well-placed. The writing of this this manuscript, however, leads me to be unclear as to where the authors' contributions lie. In particular, I believe much of the manuscript could be trimmed and cut down to reveal more clearly the fundamental notions and intuition which are missing for a reader such as myself. And given the apparent empirical benefits of this direction, I would like to see a greater variety in numerical examples. This, in combination with my principal concern that much of what the authors claim as novelty appears to me to be pre-existing, at least in spirit, in unreferenced literature and software (see Section 2.1 below), however, leads me to be unconvinced as to the manuscript's readiness for publication in its current form. I believe that many of my points below could be handled by the authors, and with some clarification and editing, this manuscript could potentially fill a gap which I feel is indeed missing in the literature and would be valuable to the readership of the Journal of the American Statistical Association and beyond.

2 Major comments

Below I list some comments, in no particular order, which fundamentally restrict my recommendation for this manuscript to be accepted for publication in its current state.

2.1 Clarity of contribution

My principal concern with the manuscript in its current state is the clarity of its novelty. In particular, the authors point out that the RSP transform is distributionally equivalent to the probability integral transform (PIT). Well, the use of PIT and the leave-one-out version of the PIT (LOO-PIT) for model diagnostics is not a novel idea: it was first discussed, to the best of my knowledge, in a reference which I would consider a somewhat critical omission from the manuscript (Gelfand et al., 1992, and in particular the discussion provided by Prof. Françoise Seillier-Moiseiwitsch). And the theoretical results the authors present seem to me to be closely related to pre-existing results mentioned therein: see, e.g., Theorem 2.1 of O'Reilly and Quesenberry (1973), Theorem 1 of Seillier-Moiseiwitsch and Dawid (1993), the review paper of Seillier-Moiseiwitsch and Seillier-Moiseiwitsch (1993). While these results are slightly different to what the authors present—namely some of them assume a natural ordering of data, though not all—it is not clear to me how different they are, and the importance of these differences in terms of the fundamental message and novelty of the manuscript at hand. Do I understand correctly that the primary contribution of the manuscript is a synthesis and extension of these previous contributions to new theoretical results which *are* missing in the literature? Or is the p-value style testing itself the authors feel is novel? The authors mention the difference in "the order of operations" (p. 12) between PPP and Z-residuals; it may be that I am confused about this difference and am referring in the above only to the former, in which case I believe a more concrete example (theoretical and numerical) would clarify this confusion. At the very least, a thorough literature review making clear what results are already available, what tools are already implemented in software, synthesising these into a unified framework (which I believe to be missing in this subfield and would make a valuable contribution in and of itself), and thus clarifying the gaps in the literature which the authors would endeavour to fill would greatly aid the manuscript.

A subsequent concern is that the use of PIT and LOO-PIT predictive diagnostics, at least in terms of testing their uniformity, is already common-place in application (a cursory search provides references such as Saleh and Rasel, 2024; Pernot, 2022, from recent years), and has already been implemented in open-source software including `ArviZ` in `Python` and `bayesplot` in `R`. As such, while the empirical promise of these tools is naturally very exciting, it does not seem to be specific to anything the authors have presented in their manuscript.

2.2 Breadth of examples

The authors present numerical demonstrations only for the negative binomial and hurdle negative binomial family. A more complete selection of model families would render these sections more convincing of the methods' empirical prowess. For instance, the authors show an example where one can diagnose the use of the wrong model family between two discrete families: could they extend this to choosing between a continuous and discrete family? For instance, in the following case study: `https://users.aalto.fi/~ave/casestudies/Nabiximols/nabiximols.html`

The authors write in the theory section that the LOO-PIT (or LOO-RSP) approach can resolve many of the problems with the standard PIT (RSP) diagnostics. Yet the real world experiment uses only RSP diagnostics and no LOO-CV equivalent. Can the authors please clarify why this is?

Further, in their Table 2, the Δ_{elpd} and se columns related to the elpd relate to the mean and standard deviation of the normal approximation of the difference in predictive performance. One could produce a summary statistic from these two columns by computing $\mathbb{P}(\Delta_{\text{elpd}} \geq 0)$ in R as

```
pnorm(0, delta_elpd, se_elpd, lower.tail = FALSE)
```

Showing this as an additional column would help articulate their conclusions.

In Section 3.2 (p. 20) the authors write that the PPC test has "significantly lower power (e.g., ~ 0.05 vs. ~ 0.4 for $n = 50$)", could the authors highlight where these numbers come from in the plot above, since it shows a great many different methods?

A small numerical example at the beginning of Section 2.8 with visualisations would be very helpful in understanding what it would mean for "a large proportion of small p-values" to represent (p. 14).

2.3 Sensitivity of test statistic and implications for multiple testing

In their numerical examples, the authors show how in some instances some tests are "more powerful" than others, and subsequently that there could exist some "well-chosen test statistic" (p. 19) for a given problem. On this point I want to offer two comments: the first is whether we can tell ahead of time which test statistics are indeed "well-chosen"; the second is, in the event that we first choose a test statistic which yields no actionable information, whether we are at risk of some multiple testing concerns by performing a new test on the same residuals. In particular, do the authors suggest that we should enumerate all possible tests we would wish to perform ahead of time? Or that we should use different hold-out folds for different tests? If not, is there some intuition as to why this does not represent a concern for type I error guarantees. Could this be mitigated by some correction? Or the use of E-values (see Section 4.1 below)?

And if we are to pre-assign a set of tests, then which tests in particular do the authors propose as reasonable defaults?

On this point, I think that the authors should make clear that while it is true that p-values based on PPCs "depend heavily on the choice of test statistic" (p. 1) the same seems also to be true for their Z-residuals. Ensuring that the language they employ does not hide this sensitivity, i.e. does not suggest that their approach entirely does away with the "subjectivity" (p. 3) of choice of test statistic, seems important to me.

In Section 3.3 the authors write that "the inclusion of noise variables in both models aims to provide a more realistic and robust testing environment" (p. 21). Could they please comment on how the form of this noise affects their conclusions? That is, if the form of the model is correct but the noise is heavy-tailed, asymmetric, or comprised of point outliers, how do the conclusions of this experiment change?

3 Minor comments

The following comments, while perhaps not fundamental to the acceptability of the manuscript, are points which I believe would improve it if considered.

3.1 Assumptions for theoretical results

In Theorem S2.2, the authors make the assumption that the model is well-specified, yet their proof technique seems only to make use of the usual conditions required for posterior concentration, which occurs independently of model well-specification (see, e.g., Example 5.25 of Vaart, 1998, in the case of a frequentist estimator, the same arguments can be easily extended to full Bayesian posteriors). As such, I am not sure where they require this assumption for their result? In particular, in Step 1 of their proof of Theorem S2.3 (which overall *does* of course require the assumption of well-specification), it may be that in the case of misspecification we would instead have

$$\|P_i^{(n),\mathrm{CV}} - P_i^{(*)}\|_{\mathrm{TV}} \to \varepsilon > 0.$$

If I have understood this correctly, could we extend their results to make this irreducible error appear in the final bound? That is, do the authors believe that it would be possible to derive a bound where the degree of misspecification, ε, is made explicit, and subsequently interpreted from a p-value?

3.2 Additional intuition

Adding some intuition for why (p. 8)

> the randomisation in the RSP is essential for ensuring a truly uniform distribution for discrete data

would have helped my understanding.

Similarly (p. 10) it is not immediately clear to me, and perhaps some other less familiar readers, why "partitioning Z-residuals into groups with respect to a covariate" is required. Is this to mitigate multiple testing concerns?

And (p. 10) when the authors mention that "Z-residuals are approximately standard normal", is this approximately present only in the importance sampling case? Or is this a pre-asymptotic statement? Again some clarification on where this bias emerges, and whether it can be quantified, would help greatly.

When discussing the adjustment to the degrees of freedom in the χ^2 statistic (p. 11) the authors adjust n (the number of data points) by p_{eff} (the effective number of parameters). Some further discussion on why this forms a valid adjustment, and the relationship between the number of data and the number of parameters, would again greatly aid comprehension.

In Section 2.6, the sentence starting "therefore" (p.11) does not follow immediately to me from what came before. Just another sentence here might help me and other readers understand more clearly.

3.3 Plotting

As mentioned above, many of the tools the authors propose seem to me to have already been implemented in software (e.g. by Gabry et al., 2019; Säilynoja et al., 2021, which could be referenced alongside Gelman et al., 2013). As such, presenting plots which are inline with pre-existing styles, or motivating the difference, would help some readers like me to understand the connection to this other, currently missing side of the literature.

In Figure 1, plotting the marginal distribution of the Z-residuals would help articulate the point the authors make.

3.4 Acronyms

While the list of abbreviations is helpful, it is also very long and indicative perhaps of an opportunity to trim them down. At some points I found myself returning to it to ensure I understood what I was reading, which distracts from the flow of the manuscript. Further, some acronyms including "QQ plots" are never introduced nor are they present in this list.

3.5 Flowery wording

The second sentence of the manuscript (p. 1)

> Statistical modelling without rigorous checking is akin to building a high-rise on sand—elegant in appearance but prone to collapse under scrutiny.

is perhaps overly romantic in style, disjointed from the rest of the paper, and distracts from the rest of the introduction in my personal opinion.

3.6 Typos and notation

1. (p. 2) the data are denoted in bold, specifying the space in which they live would help justify this
2. (p. 3) the number 0.5 comes out of thin air when discussing the bias of PPPs
3. (p. 3) defining the survival probability function here would be very helpful in introducing the concepts of the paper, as opposed to only defining is in Appendix S2.2
4. (p. 6) "((in a measure-theoretic sense)" has extra parentheses
5. (p. 8) adding a reference to where the authors formally show that "[u]nder the true model [...] Z-residuals are independent and identically distributed standard normal variables" would help
6. (p. 9) the notation around Equation 5 could be refined, it is currently quite heavy
7. (p. 15) "(supplementary))" could be deleted, and has extra parentheses

4 Curiosity

Below I list some questions which are perhaps tangential to the ambition of the manuscript and come purely from my own personal curiosity. The authors need not feel they should adjust the manuscript to answer them, but if they have any thoughts on them I would appreciate whatever response they have.

4.1 E-values and multiple predictive testing

Is there space for E-values (Grünwald et al., 2024) to replace the p-value-style tests the authors propose here? Do the authors believe this could resolve my question of multiple testing in Section 2.3?

4.2 Rates of convergence

The authors make the point at the beginning of Section 2.7 that

> for both the cross-validated and posterior constructions, the RSPs are asymptotically independent and uniformly distributed.

Do we have any understanding of whether this occurs at the same rate for both constructions? And subsequently, at what rate they are indistinguishable from one another?

4.3 Beyond perfect model specification

Related to Section 3.1 above, the authors have shown their theoretical results under correct model specification. While this is clearly the standard in the literature, I'm sure the authors will agree that perfect model specification is a fantasy in reality. Do they believe that the sorts of results they have developed (e.g. Theorem S2.3 *as well as* Theorem S2.2), and those pre-existing in the literature could be extended to notions of ε-misspecification I previously detailed? That is, where the model is misspecified, but perhaps not-so-gravely misspecified as to make a noticeable difference in reality?

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